AI-Native RAN: 6G Day 1 Operator

Towards AI-Native RAN — China Mobile, arXiv:2507.08403

2026-06-24

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China Mobile(Nan Li, Qi Sun, Chih-Lin I) **operator 6G AI-Native RAN position** . , “6G release(Day 1) AI native ” 2G~5G · . .

- 5G AI/ML **add-on** , legacy architecture · protocol · flexibility . 6G AI native .
- **AI for RAN**(AI RAN) **RAN for AI**(RAN pervasive AI compute/connectivity) .

(1) AI-Native RAN essential capability , (2) **AI Node** RAN node Day 1 , (3) 5000+ 5G-A field trial .

(arXiv:2507.08403v1) .

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2 Motivation: add-on Day 1

2.1 native

6G RAN AI . (i) massive MIMO signal processing, (ii) UE · topology radio resource allocation scheduling · . real-time , AI .

2.2 Day 1

release . release 1.5~2 , operator TCO . 4G/5G release(Rel-8, Rel-15) , release vendor , operator .

> **(HST)** . HST 2024 48,000 km, 350 km/h , Doppler shift Rel-16 live network . 6G , AI Day 1 “AI-as-afterthought” 6G .

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3 essential capability

AI-Native RAN capability (Fig. 1).

3.1 Capability 1 — AI-Driven RAN Processing / Optimization / Automation

L1/L2/L3 AI . AI : , · real-time , · proactive .

Table I , use case .

- **AI for resources control & optimization (L2/L3)**: load balancing, mobility management, network energy saving, network slicing, capacity & coverage optimization, **multi-user scheduling & resource allocation**, QoS/QoE & SLA assurance.
- **AI for air interface (L1)**: positioning, beam management, CSI compression/prediction, AI receiver/transceiver, PA non-linearity compensation, interference prediction, RS overhead reduction.

use case inference (BS/UE/LMF-side), model · (DNN/LSTM/DRL/GNN/Transformer, `<1M~10M-1B`), status(3GPP Rel-18/19, O-RAN, *6G candidate use case*) .

> . `multi-user scheduling and resources allocation` **6G candidate use case** , `DNN(<1M)`, `DRL(<10M)` . scheduling "AI " , utility (→ 7).

3.2 Capability 2 — · · AI LCM

· · .

(a) Data collection. 5G RAN protocol AI . control plane payload , user plane core UPF charging RAN AI , radio link connectivity . :

- *Dedicated AI radio bearer*: RAN · bearer training data · model · metric . RAN node AI data tunnel .
- *Cross-domain data collection*: application server traffic service QoE RAN . LLM inference **TTFT(time to first token)** PDU set mapping · DRB configuration .
- *Task-driven customizable collection*: 15 OAM **10 ms** latency , UE , filter (: `UE velocity > 250 km/h`) .

(b) Model management. off-line mismatch online · fine-tuning pre-validation . monitoring **model performance / network performance / resource utilization** 3 (Table II). UE model context(ID ·) , performance-triggered/periodic retraining, RAN Digital Twin pre-deployment , FL · large foundation model .

(c) Distributed collaborative computing. node compute · energy · AI node . collaborative *training / inference / fine-tuning* , gNB · UE compute capability logical compute pool resource pooling · dynamic scheduling .

3.3 Capability 3 — AIaaS Provisioning

operator connectivity → edge AI compute → edge AI service enabler . service mode(Fig. 8):

- **Connection assurance for AI services**: multimodal · model update flow QoS framework , **traffic pattern** (embodied intelligence lateral flow, personal assistant *uplink-heavy · bursty* traffic), **deterministic latency**(assistant `< 200 ms`, embodied AI `< 100 ms`) .
- **Data services**: traffic load · link quality metric mobility · location sensing . robotic system sensing/update packet loss .
- **RAN AI computing services**: GPU/NPU compute edge AI (- joint scheduling).

open API framework .

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4 Day 1

1. **Day 1** — (HST).
2. **Forward compatibility** — (hardware) release , centralized resource management + distributed processing site idle compute . (software/model) AI model processing block · protocol layer .
3. **RAN AI service exposure** — AI compute , idle AI capacity cloud AI connectivity .
4. **3GPP + SDO** — 3GPP end-to-end · · interoperability , ITU umbrella framework/ (FG-AINN), ETSI · O-RAN RAN cloud · RAN application(API) . AI Node 3GPP , O-RAN .

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5 : AI Node + 6gNB

(Fig. 9). **timeliness** .

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RAN OAM	central cloud, non-real-time	supervision · coordination, MDA AI

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AI Node	locally centralized, AI compute(edge platform)	data collection, model management, **model repository**, **AI/ML training**(L1/L2/L3), **L3 inference**, RAN capability exposure
6gNB	distributed, UE	(dynamic scheduling) + **L1/L2 real-time inference** + LCM/
6G UE		+ ( ) inference ·

AI Node 6gNB **one-to-many cardinality** , RAN interface .

:

- **training AI Node** → 6gNB AI compute , sporadic training **statistical multiplexing** (AI BS inference →).
- **real-time L1/L2 inference 6gNB .** functional split "L1/L2 real-time " .
- compute OTT/RAN App AIaaS (new SBI core , OAM).

timeliness split "L1/L2 real-time node-local " . Near-RT RIC xApp externalize node collocate (→ 7).

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6 Field Trial

China Mobile 2024 3 trial. AI Node **CCU(Centralized Computing Unit)** .

- **Phase 1** (2024.1~12, **31 · 5000 5G-A**): - AI + use case .
- **Phase 2**: 475,000 gNB , · , AI RRM, experience-centric .
- **Phase 3**: , UAV mobility management, embodied intelligence QoE, RAN capability exposure(AIaaS).

6.1 Case 1 —

AI traffic (1000+ app type, 95% ; 8 gNB 4,800) , QoS trigger . **utility** :

$$\hat{\mathbf{a}}_u \;=\; \arg\max_{\mathbf{a}\in\mathcal{A}} \; \mathbb{E}[\,U(\hat{\mathbf{q}}_u(\mathbf{a},\mathbf{x}_u))]$$

- $\mathbf{a}$: u action(scheduling weight, resource allocation)
- $\mathbf{x}_u$: context(application type, RAN metric, mobility state)
- $\hat{\mathbf{q}}_u(\cdot)$: AI **service-level quality** (latency, throughput, QoE)
- $U(\cdot)$: operator policy utility function(QoE latency)

: poor coverage(RSRP ≤ -105 dBm) · heavy load(PRB $\geq 50\%$) short video latency **33.6% / 30.5%** , **43.0 ms → 32.0 ms (25.6%)**. QR code **18.5 ms → 14.5 ms (21.9%)**. link 720p→1080p/4K **3~6% traffic** .

6.2 Case 2 — root cause

UE · network · application / :

$$\hat{y}_u \;=\; f_\theta\big(\mathbf{x}_u^{\text{App}},\, \mathbf{x}_u^{\text{RAN}},\, \mathbf{x}_u^{\text{UE}},\, \mathbf{x}_u^{\text{Mob}}\big)$$

App (TCP RTT · packet loss), RAN(SINR · RSRP · handover), UE , mobility(grid positioning, $50\,m \times 50\,m$) **XGBoost** user type(indoor/outdoor/high-speed) **90%** **precision** . rule **+20%**, recall **+30%**.

6.3 Case 3 — network energy saving

· · · 20 ES , :

$$\mathbb{E} = \sum_{n=0}^{N}\sum_{m=0}^{M}\sum_{t=0}^{T}\Big[P^{PA}(P,m,t)+P^{\text{transceiver}}(m,t)+P^{\text{digital IF}}(m,t,C)\Big]+\sum_{n=0}^{N}\sum_{t=0}^{T}\Big[P^{\text{baseband}}(t,C)+P_0\Big]$$

ES · traffic . AI :

- *Solution 1*: NMS AI 15 cell-level traffic ES → **26.6%** .
- *Solution 2*: BS CCU service type , delay-tolerant traffic packet aggregation · sleep → **34.16%** .

(AI +5.32%, +12.88%.)

6.4 Trial

CCU **multi-vendor gNB** cross-platform collaborative inference , AI **API** · . → Day 1 (interoperable AI service framework) .

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7 gap thesis

"L2 scheduler token importance traffic utility " . importance-aware MAC scheduling .

( )	(thesis gap)
Case 1 **utility** scheduling ($\arg\max_{\mathbf{a}} \mathbb{E}[U(\hat{\mathbf{q}}_u(\mathbf{a}, \mathbf{x}_u))]$) — operator	$U(\cdot)$ $\hat{\mathbf{q}}_u$ **token importance(model entropy · attention)** utility → end-to-end task utility/resource
Table I multi-user scheduling **6G candidate use case** (DNN/DRL,)	PRB/TB granularity **importance-aware L2 scheduling** utility · (BLER task utility)
Connection assurance: embodied intelligence lateral flow, UL-heavy · bursty traffic, deterministic latency(`<100 ms` embodied)	(stationary arm/AMR/drone) **mobility** offloading · UL scheduling → motivation
Cross-domain data collection: LLM **TTFT** → PDU set mapping · DRB	token content **predictive resource demand estimation**(Stage 3)
Task-driven 10 ms data collection, filter(: velocity>250 km/h)	**proactive PRB pre-allocation(chance-constrained, Stage 4)**
Timeliness functional split: L1/L2 real-time inference 6gNB(node-local)	"sub-TTI scheduling GPU-collocate Near-RT RIC xApp externalize "

7.1

- **citation** : China Mobile · Chih-Lin I operator/ → AI-RAN .
- **motivation** : UL-heavy · bursty traffic, robot offloading, deterministic latency use case .
- **gap** : "scheduling utility " (form) , utility **(token importance)**, · . Stage 3(demand estimation) · Stage 4(proactive pre-allocation) .

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- : *Towards AI-Native RAN: An Operator's Perspective of 6G Day 1 Standardization*
- : Nan Li, Qi Sun, Lehan Wang, Xiaofei Xu, Jinri Huang, Chunhui Liu, Jing Gao, Yuhong Huang, Chih-Lin I (China Mobile)
- arXiv: 2507.08403 (cs.NI), 2025-07-11